COMPASS: TEMPEST Discrete DOC Data QAQC

March 2026

2026-03-20

Run Information

```
#identify which section you are in
cat("Run Information")
```

```
## Run Information
```

```
#a link to the Gitbook or whatever protocol you are using for this analysis
  #steph will add this soon

#anything that needs to be changed do this in the first chunk
  Date_Run = "03/18/26"
  Run_by = "Stephanie J. Wilson"
  Script_run_by = "Stephanie J. Wilson"
  run_notes = "Check standards started going out of range after the samples were run.
              Looked like the flow of the instrument was disrupted, but this was after the
              samples ran, so I think it is okay to use these data."

  #file path and name for summary file
    raw_file_name = "tmp_doc_raw_data_2026/TMP_202603.txt"
  #file path and name for the all peaks file
    raw_allpeaks_name = "tmp_doc_raw_data_2026/TMP_202603_allpeaks.txt"
  #file path and name for processed data after QAQC
    processed_file_name = "tmp_doc_processed_data_2026/TMP_PW_DOC_Processed_202603.csv"

#check standard concentrations - Update if running different checks:
   chk_std_c = 50
   chk_std_n = 2

#Log path
    Log_path = "tmp_doc_raw_data_2026/COMPASS_TMP_TOCTN_QAQClog_2026.csv"
```

Setup

Pull in active porewater tracking inventory sheet

```
## File already exists. No download needed.
```

Import Data Functions

Import Sample Data

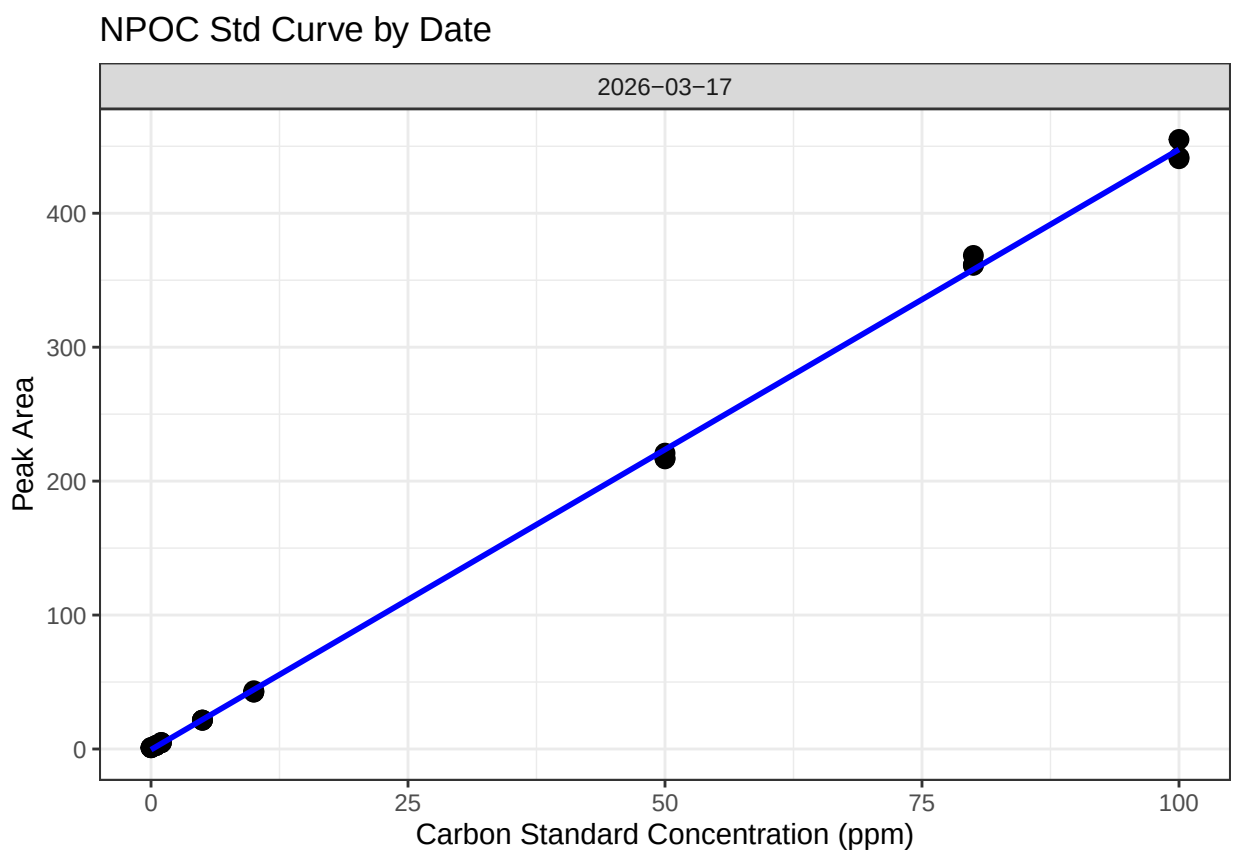
```
## Import Sample Data
```

```
## New names:
## * `` -> `...14`
```

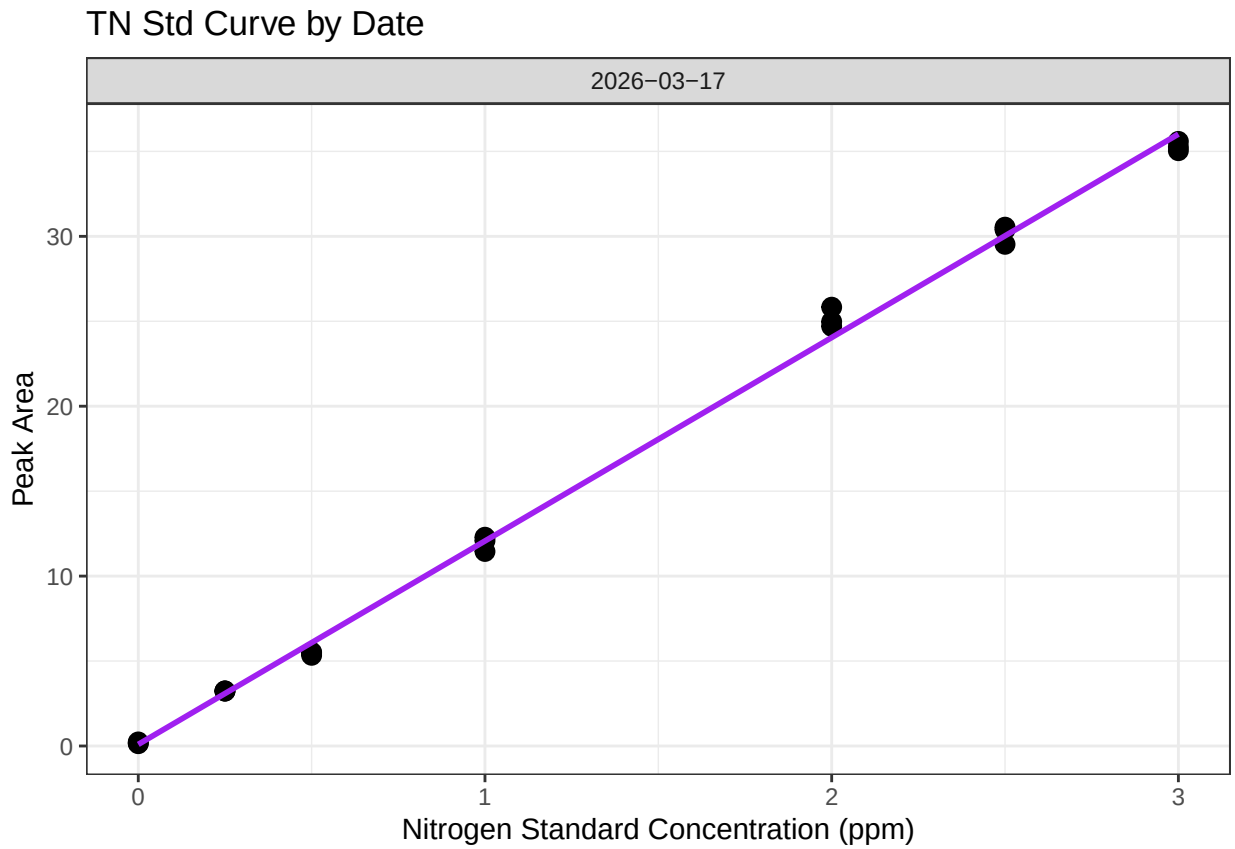
```
## # A tibble: 6 x 4
##   sample_name       npoc_raw tdn_raw run_datetime
##   <chr>                <dbl>   <dbl> <chr>
## 1 TMP_C_B4_20260309     17.2   0.333 3/18/2026 7:52:08 AM
## 2 TMP_C_C3_20260309     25.7   0.505 3/18/2026 8:16:42 AM
## 3 TMP_C_D5_20260309      8.9   0.223 3/18/2026 8:47:47 AM
## 4 TMP_C_F4_20260309     22.1   0.545 3/18/2026 9:30:20 AM
## 5 TMP_C_F6_20260309     34.0   0.690 3/18/2026 10:00:49 AM
## 6 TMP_C_H3_20260309     29.0   0.930 3/18/2026 10:31:56 AM
```

Assessing standard Curves

```
## Assess the Standard Curve

## New names:
## * `` -> `...18`

## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)

## `geom_smooth()` using formula = 'y ~ x'
```

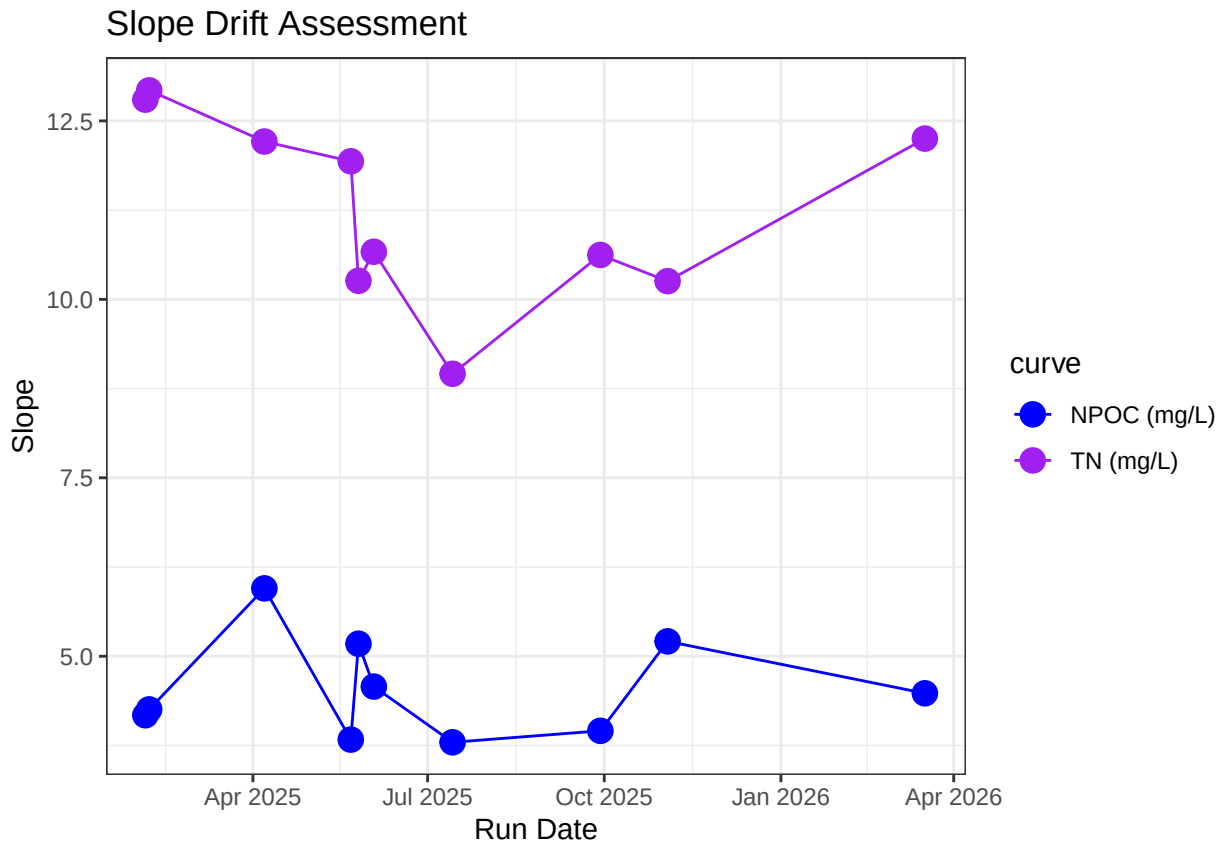



```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```

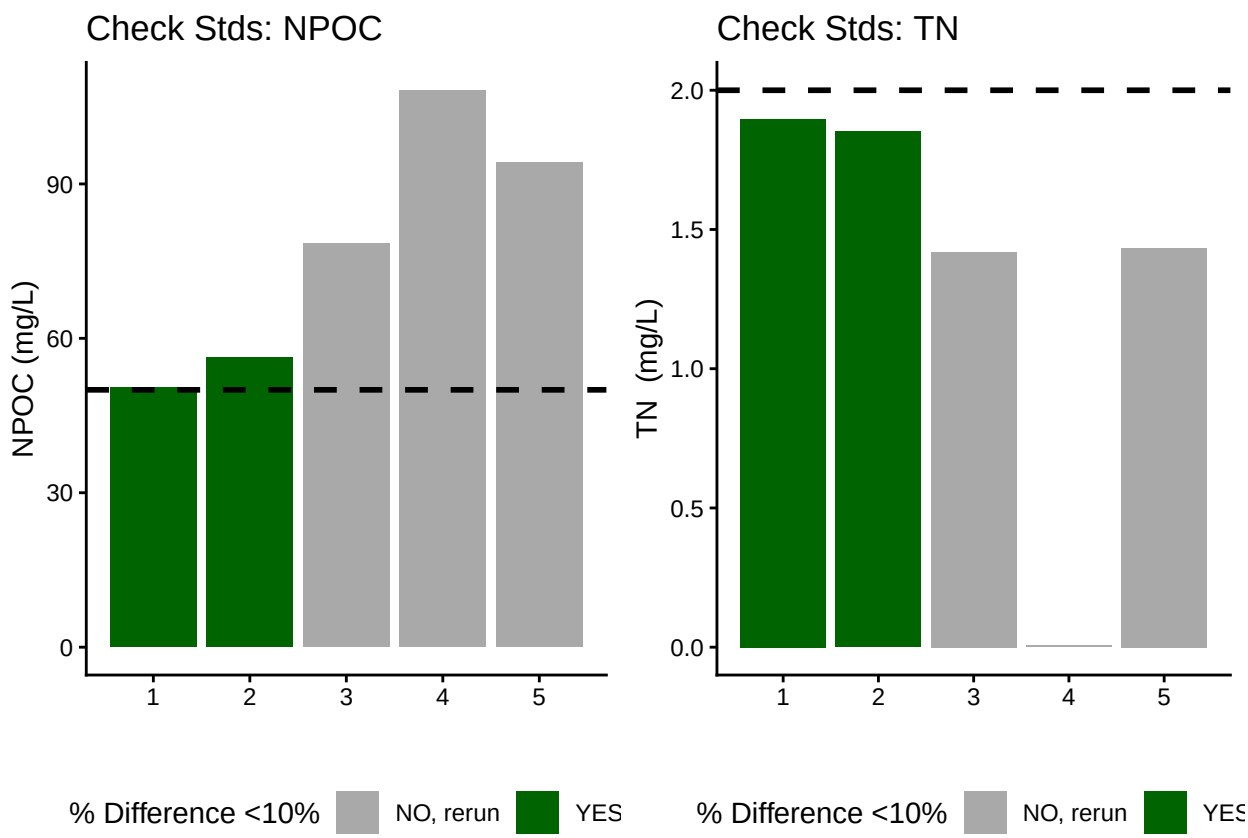



```
## [1] "NPOC Curve r2 GOOD"
```

```
## [1] "TN Curve r2 GOOD"
```

Assess Check Standards

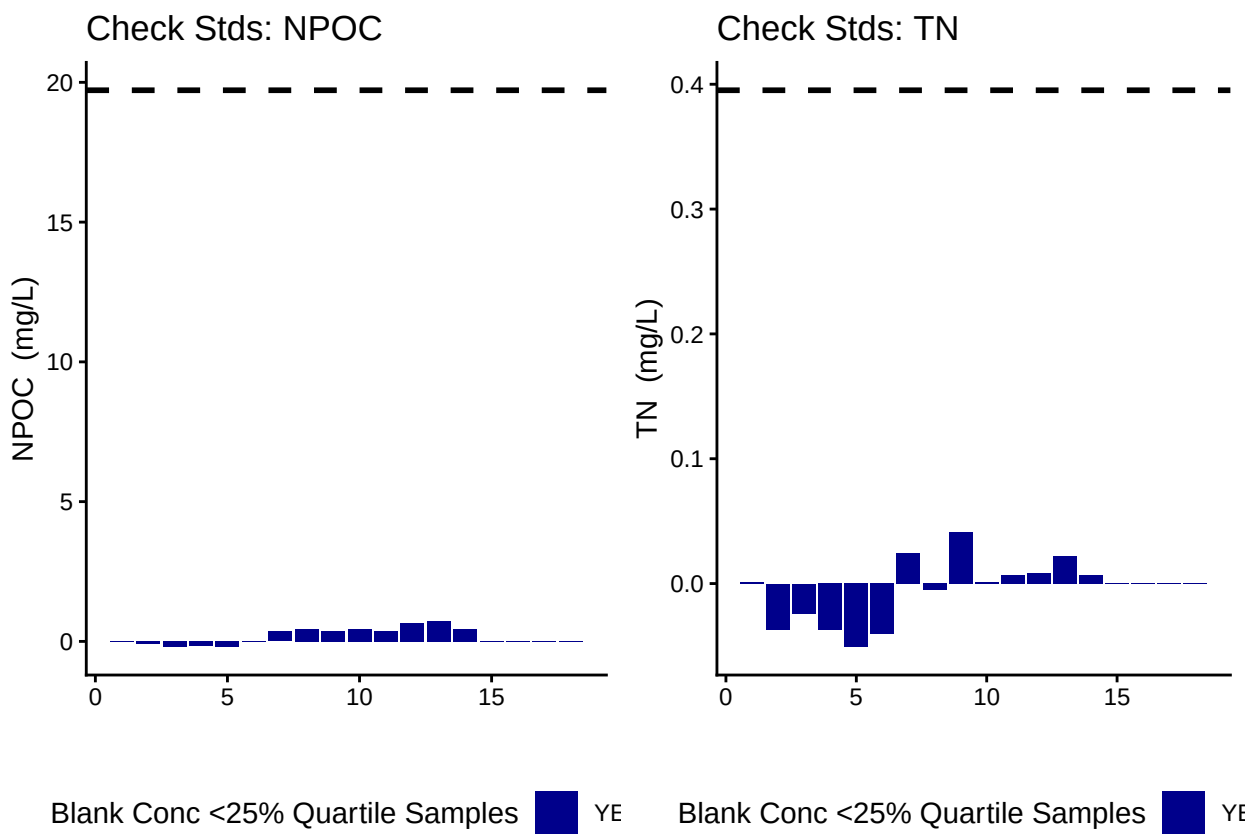
```
## Assess the Check Standards

## New names:
## * `` -> `...14`

## [1] "Carbon CHECK STANDARD RSD TOO HIGH - REASSESS"

## [1] "Nitrogen CHECK STANDARD RSD TOO HIGH - REASSESS"
```




```
## [1] "<60% of Carbon Check Standards are within range of the expected concentration - REASSESS"

## [1] "<60% of Nitrogen Check Standards are within range of the expected concentration - REASSESS"
```

Assess Blanks

```
## Assess Blanks

## New names:
## * `` -> `...14`

## [1] ">60% of Carbon Blank concentrations are below the lower 25% quartile of samples"

## [1] ">60% of Nitrogen Blank concentrations are below the lower 25% quartile of samples"
```

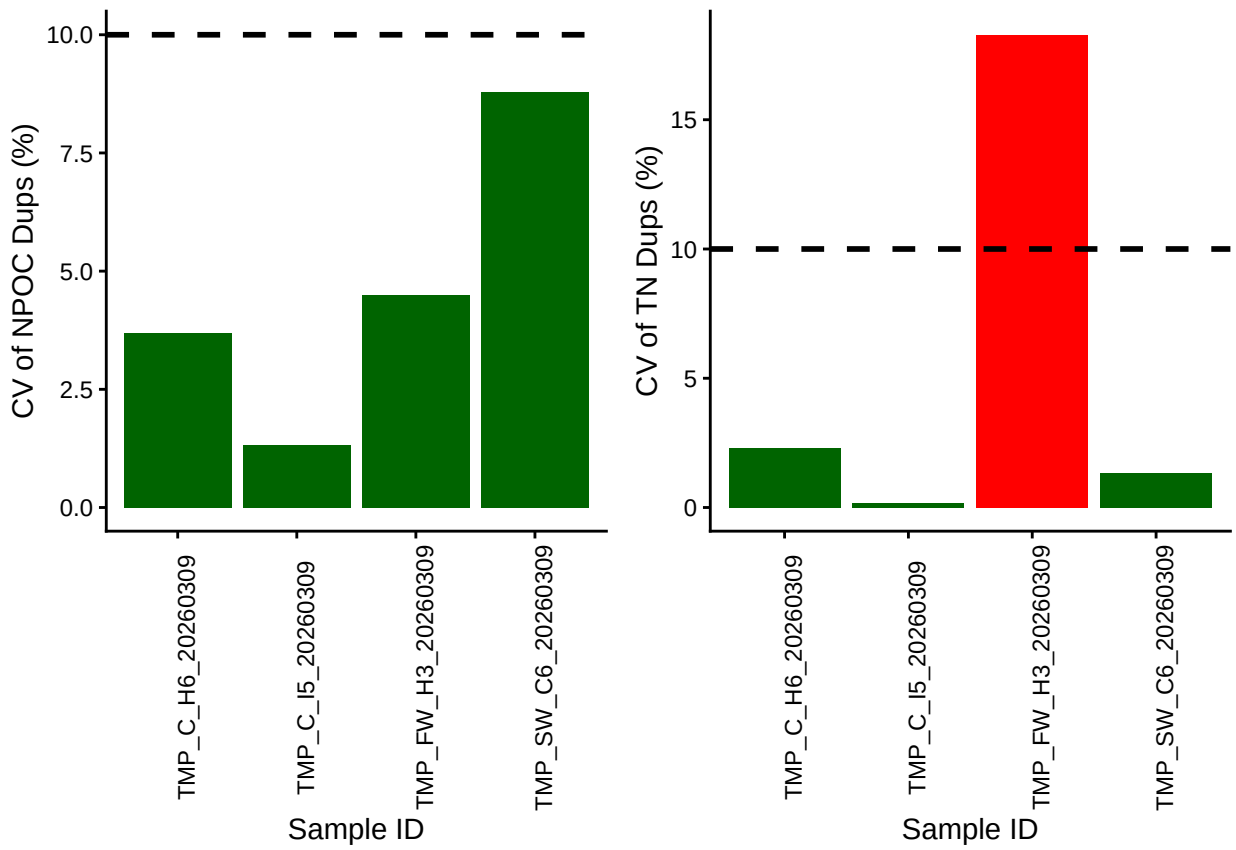



```
## carbon blanks:

## [1] 0.1713417

## nitrogen blanks:

## [1] -0.004562222
```

Assess Duplicates

```
## Assess Duplicates
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

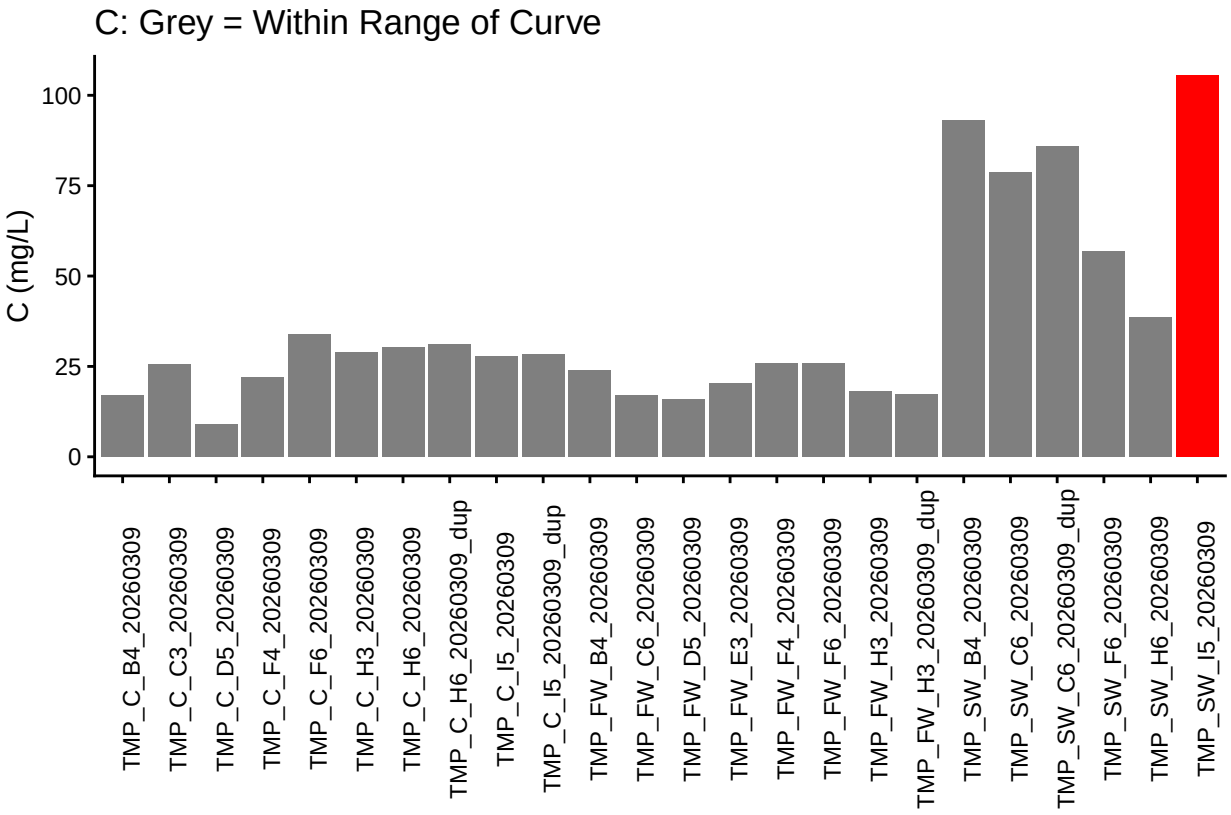


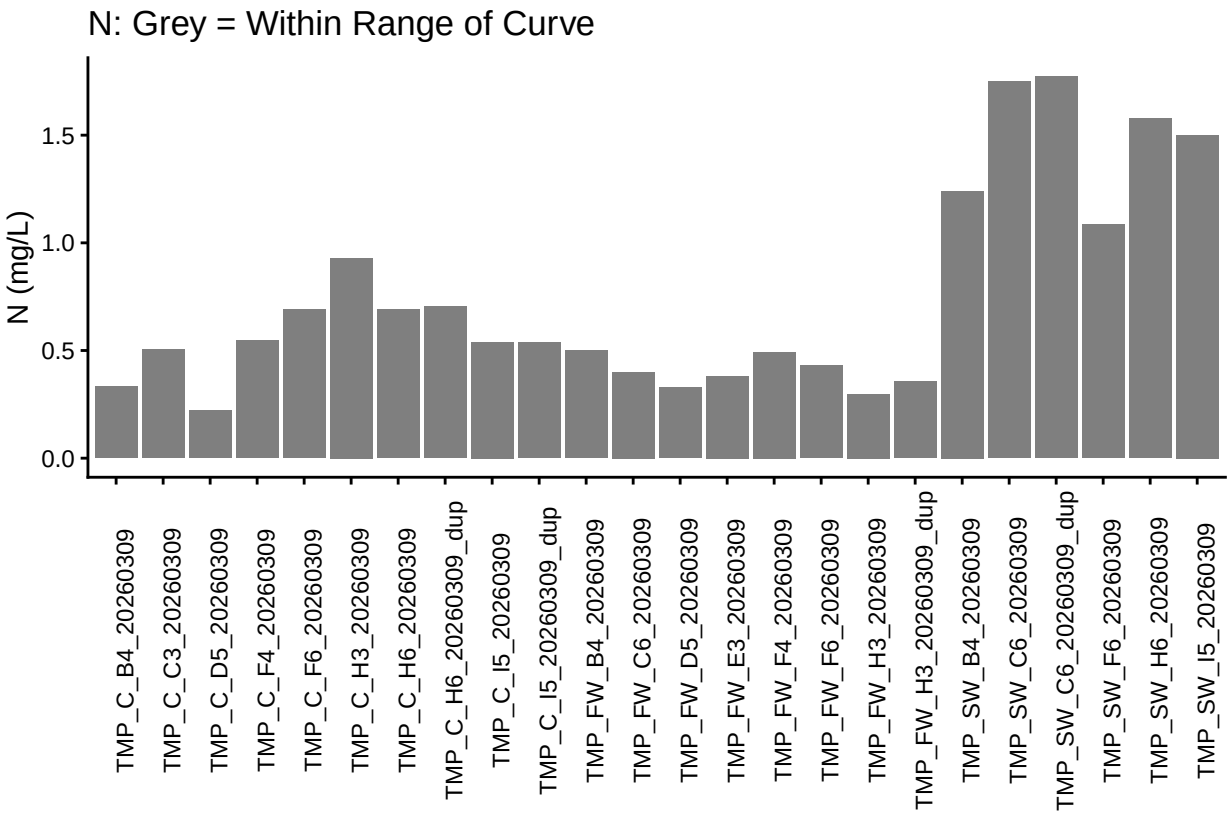

```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

```
## [1] ">60% of Nitrogen Duplicates have a CV <10%"
```

Sample Flagging

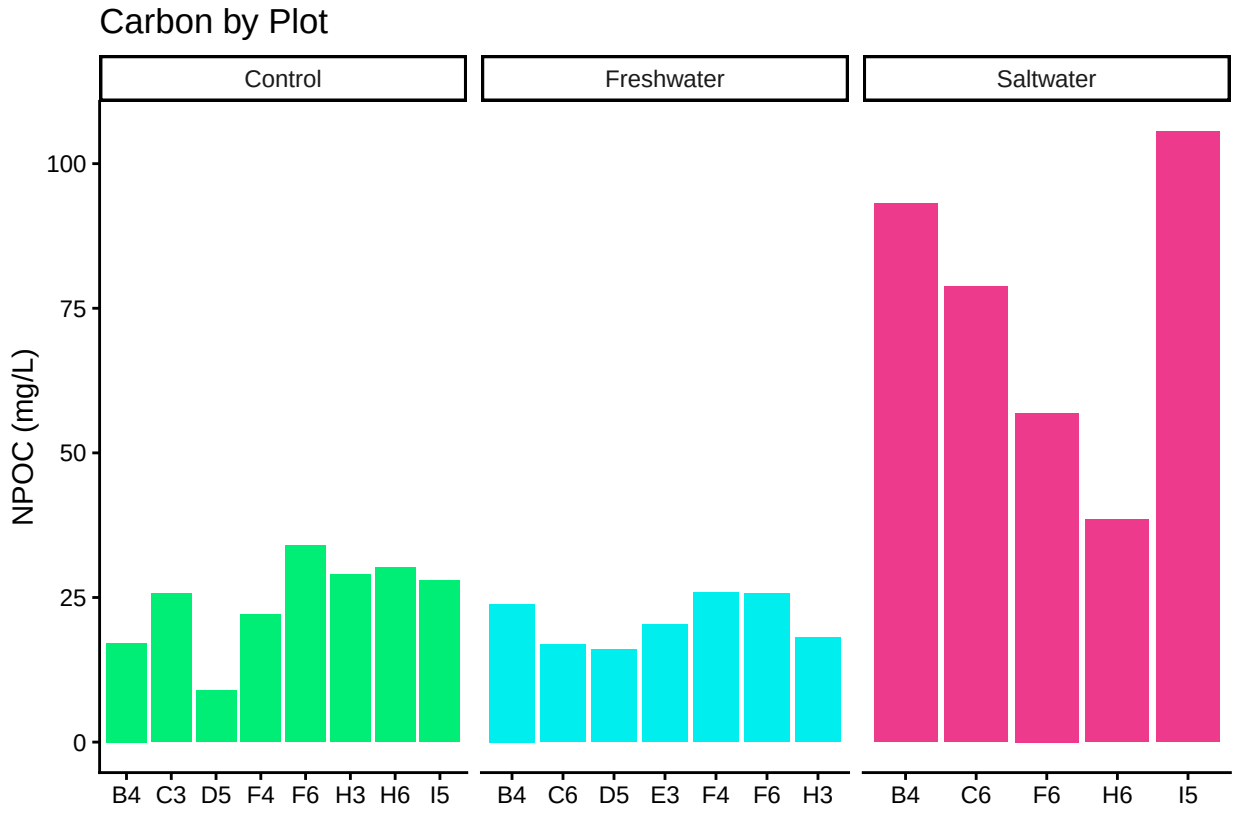
```
## Sample Flagging
```

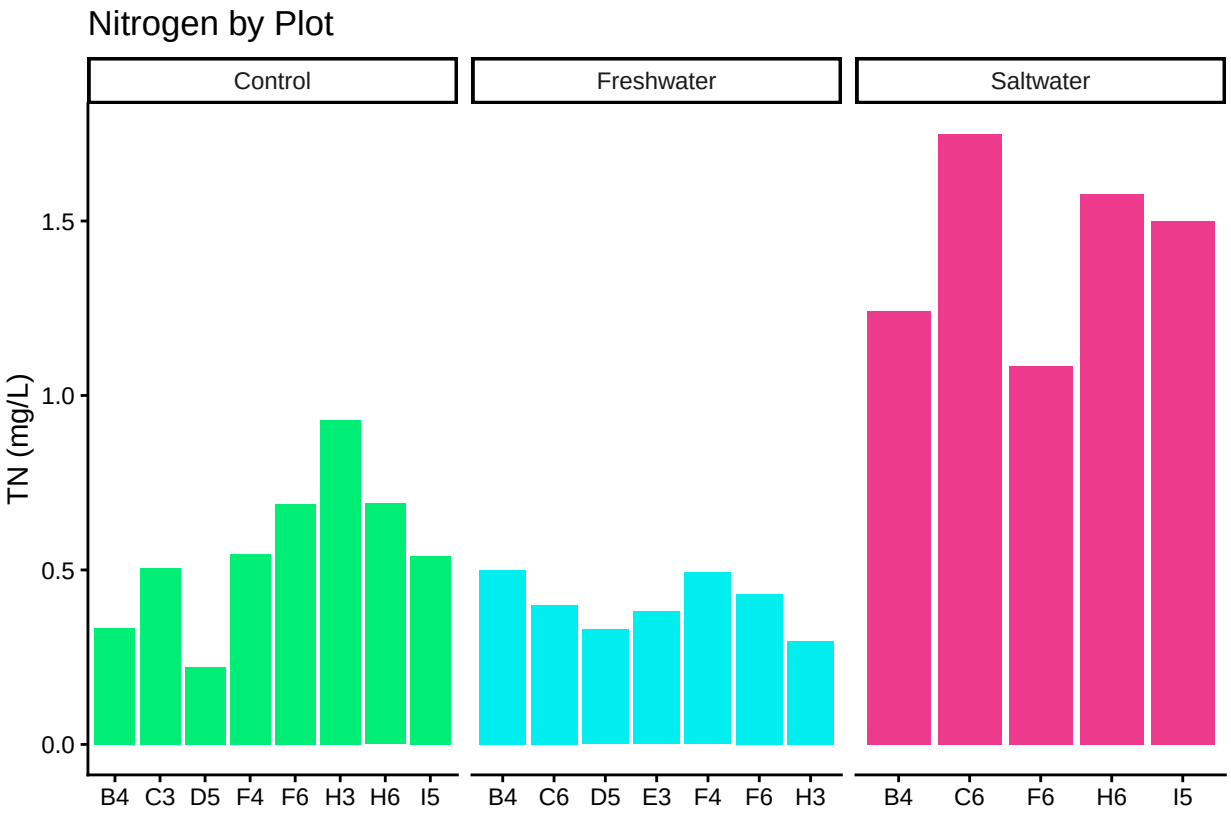



N: Grey = Within Range of Curve
N (mg/L)
0.0
0.5
1.0
1.5
TMP_C_B4_20260309
TMP_C_C3_20260309
TMP_C_D5_20260309
TMP_C_F4_20260309
TMP_C_F6_20260309
TMP_C_H3_20260309
TMP_C_H6_20260309
TMP_C_H6_20260309_dup
TMP_C_I5_20260309
TMP_C_I5_20260309_dup
TMP_FW_B4_20260309
TMP_FW_C6_20260309
TMP_FW_D5_20260309
TMP_FW_E3_20260309
TMP_FW_F4_20260309
TMP_FW_F6_20260309
TMP_FW_H3_20260309
TMP_FW_H3_20260309_dup
TMP_SW_B4_20260309
TMP_SW_C6_20260309
TMP_SW_C6_20260309_dup
TMP_SW_F6_20260309
TMP_SW_H6_20260309
TMP_SW_I5_20260309

Visualize Data by Plot

```
## Visualize Data
```




Nitrogen by Plot
Control
Freshwater
Saltwater
TN (mg/L)
0.0
0.5
1.0
1.5
B4 C3 D5 F4 F6 H3 H6 I5
B4 C6 D5 E3 F4 F6 H3
B4 C6 F6 H6 I5

Convert data from mg/L to uMoles/L

Add in/check metadata

```
## Check Sample IDs with Metadata
```

```
## # A tibble: 20 x 2
##    sample_name        metadata_recorded
##    <chr>              <lgl>
##  1 TMP_C_B4_20260309  TRUE
##  2 TMP_C_C3_20260309  TRUE
##  3 TMP_C_D5_20260309  TRUE
##  4 TMP_C_F4_20260309  TRUE
##  5 TMP_C_F6_20260309  TRUE
##  6 TMP_C_H3_20260309  TRUE
##  7 TMP_C_H6_20260309  TRUE
##  8 TMP_C_I5_20260309  TRUE
##  9 TMP_FW_B4_20260309 TRUE
## 10 TMP_FW_C6_20260309 TRUE
## 11 TMP_FW_D5_20260309 TRUE
## 12 TMP_FW_E3_20260309 TRUE
## 13 TMP_FW_F4_20260309 TRUE
## 14 TMP_FW_F6_20260309 TRUE
## 15 TMP_FW_H3_20260309 TRUE
## 16 TMP_SW_B4_20260309 TRUE
## 17 TMP_SW_C6_20260309 TRUE
## 18 TMP_SW_F6_20260309 TRUE
## 19 TMP_SW_H6_20260309 TRUE
## 20 TMP_SW_I5_20260309 TRUE
```

Export Processed Data

```
## Export Processed Data
```

```
## # A tibble: 6 x 22
##   Project        plot  grid  Depth_cm sample_type Vial_ID date  npoc_mgL npoc_uM
##   <chr>          <chr> <chr>    <dbl> <chr>       <chr>   <chr>    <dbl>   <dbl>
## 1 COMPASS: TEMP~ C     B4          15 DOC         C_B4_D~ 2026~     17.2   1429.
## 2 COMPASS: TEMP~ C     C3          15 DOC         C_C3_D~ 2026~     25.7   2142.
## 3 COMPASS: TEMP~ C     D5          15 DOC         C_D5_D~ 2026~      8.9    742.
## 4 COMPASS: TEMP~ C     F4          15 DOC         C_F4_D~ 2026~     22.1   1842.
## 5 COMPASS: TEMP~ C     F6          15 DOC         C_F6_D~ 2026~     34.0   2832.
## 6 COMPASS: TEMP~ C     H3          15 DOC         C_H3_D~ 2026~     29.0   2413.
## # i 13 more variables: npoc_flag <chr>, tdn_mgL <dbl>, tdn_uM <dbl>,
## #   tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>, Field_notes <chr>,
## #   Evacuation_date_YYYMMDD <dbl>, Collection_Date_YYYYMMDD <dbl>,
## #   Collection_Start_Time_24hrs <dbl>, Collection_End_Time_24hrs <dbl>,
## #   EST_EDT <chr>, Volume_mL <dbl>
```

#end